

Appendix A

Determination of Transformation and Position Circles

A.1 INTRODUCTION

The input impedance Z_{in} of a transmission line with a given characteristic impedance Z_1 and length d_1 and terminated in a load impedance Z_L (as shown in the insert of Fig. A.1) is usually determined either by use of an appropriate computer program or by use of a Smith chart. In view of the fact that the computer program is usually faster and more accurate than the Smith chart, one is entitled to pose the simple question, “Why consider a graphical approach at all?” The answer is simple: To guide us in our design procedure by better pinpointing where the problems are and, in particular, which components should be used or avoided. Illustrations of these concepts are given in Chapter 6 and Appendix B.

The Smith chart is well-suited for handling transmission line problems when normalized to the characteristic impedance of the transmission line in question. However, when working with several transmission line sections with different characteristic impedances, for example, we must renormalize the Smith chart each time a change is made to a new characteristic impedance. Since all Smith charts “look alike” after normalization, we can easily lose track of just exactly where we are when dealing with a complex matching problem.

One way to overcome this shortcoming is to stay in the same classical rectangular coordinate system regardless of the characteristic impedance. Although this leads to *transformation circles* that are no longer concentric and *position circles* that are no longer straight lines through the center of the normalized Smith chart, the advantage is that an entire complex problem can be kept on *one* single piece

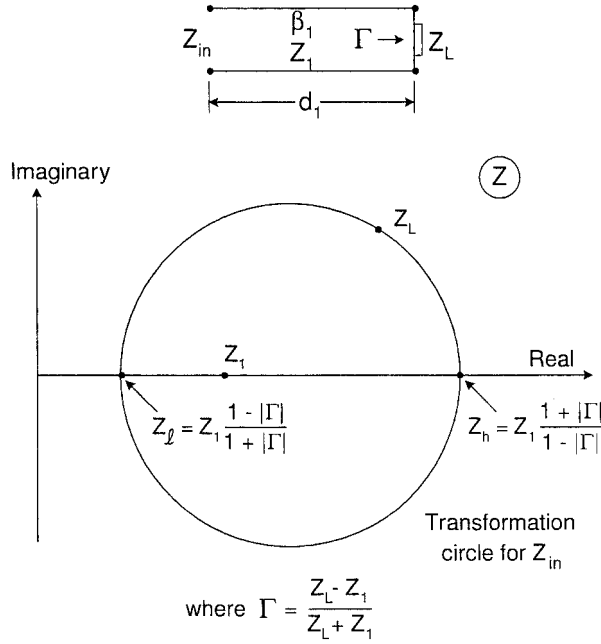


Fig. A.1 Case I: Given the load impedance Z_L and characteristic impedance Z_1 , numerically calculate Γ , Z_l and Z_h from the formulas in the figure. This completely determines the transformation circle going through Z_L . Note: This approach is valid for Smith charts as well, as long as we read all the impedances in the Smith chart.

of paper. The next step is that this technique can also be used in a single Smith chart normalized to whatever impedance one prefers. It is emphasized that the approach is recommended primarily to develop a concept. The actual calculations should be executed on a computer.

A.2 CASES DEMONSTRATING HOW TO CONSTRUCT TRANSFORMATION CIRCLES

In the following, several cases will demonstrate how to construct the transformation circles given various parameters.

Case I: Given Load Impedance Z_L and Characteristic Impedance Z_1

We seek the input impedance Z_{in} of a transmission line with length d_1 , characteristic impedance Z_1 , and propagation constant β_1 , and that terminates at an arbitrary load impedance Z_L (as shown in the insert of Fig. A.1).

It is shown in reference 122 that

$$Z_{in} = Z_1 \frac{1 + \Gamma e^{-j2\beta_1 d_1}}{1 - \Gamma e^{-j2\beta_1 d_1}}, \quad (\text{A.1})$$

where the reflection coefficient Γ is given by

$$\Gamma = \frac{Z_L - Z_1}{Z_L + Z_1}. \quad (\text{A.2})$$

Furthermore, it is shown in the same reference that Z_{in} will be located on a circle determined by the two extreme points

$$Z_h = Z_1 \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad \text{and} \quad Z_l = Z_1 \frac{1 - |\Gamma|}{1 + |\Gamma|}, \quad (\text{A.3})$$

as shown in Fig. A.1.

If the load impedance Z_L and the characteristic impedance Z_1 are known numerically (i.e., as simple complex numbers), one can readily determine Γ from (A.2) and subsequently Z_h and Z_l from (A.3). Thus, the locus circle for Z_{in} is readily given by Z_h and Z_l as shown in Fig. A.1. However, there are cases when Z_L as well as Z_1 are merely known by their geometric location in the complex plane.

Thus we will next show how to determine the locus circle for Z_{in} by purely geometrical means. *Note:* This approach holds for Smith charts as well, as long as all the impedances are read in the Smith chart.

Case II: Both Z_L and Z_1 Located on the Real Axis

Here the locus circle for Z_{in} is sought when both Z_L and Z_1 are located on the real axis, as shown in Fig. A.2.

Solution: Draw a circle with center at the origin O and radius $|Z_1|$. Next, draw the tangent from Z_L to this circle touching at point A . Draw a line from A that is orthogonal to the real axis which determines the intersection point B . Point B is the point Z_l that together with Z_h ($Z_L = Z_h$) determines the locus circle.

Proof: The two triangles OAB and OB_1A are similar. Thus,

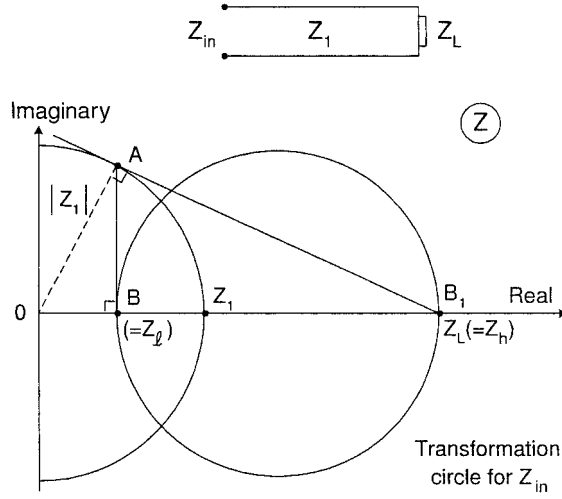
$$\frac{OB}{OA} = \frac{OA}{OB_1}$$

or

$$OB \cdot OB_1 = (OA)^2 = |Z_1|^2; \quad (\text{A.4})$$

that is, B is harmonic related to B_1 with $OB_1 = Z_L = Z_h$ and therefore $OB = Z_l$. The locus circle for Z_{in} is simply drawn through Z_l and Z_h .

Note: This approach is valid for only the rectangular coordinate system.



From the similar triangles OAB and OB_1A : $\frac{OB}{OA} = \frac{OA}{OB_1}$ or $Z_l \cdot Z_h = |Z_1|^2$.

Fig. A.2 Case II: Given the load impedance Z_L and the characteristic impedance Z_1 as points in the complex plane (both real). The low impedance Z_l is obtained by drawing the tangent from Z_L to the circle $z = |Z_1|e^{j\phi}$. Next, project the touching point A upon the real axis to obtain point B , which determines Z_l . The points Z_l and $Z_h = Z_L$ completely determine the transformation circle going through Z_L . Note: Only valid for the rectangular coordinate system.

Case III: Load Impedance Z_L Anywhere in the Complex Plane. Characteristic Impedance Z_1 on the Real Axis

For the third case, we seek the locus circle for Z_{in} when the load impedance Z_L is located anywhere in the complex plane and the characteristic impedance Z_1 is on the real axis, as shown in Fig. A.3.

Solution: Draw a circle with center at the origin O and radius $|Z_1|$. Draw the tangent from Z_L to that circle touching at point A . Next, draw a line from A orthogonal to the line through O and Z_L to determine the intersection point B . The locus point for Z_{in} is the circle going through B and Z_L with its center on the real axis (we assume Z_1 is real). The same circle intersects the real axis at points C and C_1 when $OC = Z_l$ and $OC_1 = Z_h$.

Proof: The two triangles OAB and OB_1A are similar. Thus,

$$\frac{OB}{OA} = \frac{OA}{OB_1}$$

or

$$OB \cdot OB_1 = (OA)^2 = |Z_1|^2.$$

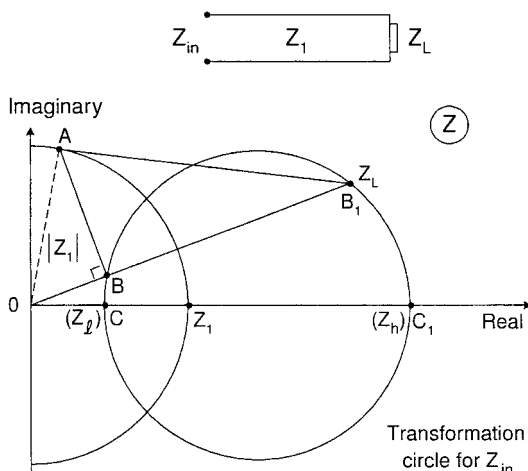


Fig. A.3 Case III: Given Z_L (arbitrary) and the characteristic impedance Z_1 on the real axis. Draw a tangent from Z_L to the circle $z = |Z_1| e^{j\phi}$ and project it onto the line OB_1 producing point B . The transformation circle for Z_{in} will go through B and B_1 with center on the real axis (for Z_1 real). This approach is only valid in the rectangular coordinate system.

Furthermore, from the “Power of the Circle” [123] we also have

$$OC \cdot OC_1 = OB \cdot OB_1, \quad (\text{A.5})$$

so by using (A.4) we obtain

$$OC \cdot OC_1 = |Z_1|^2; \quad (\text{A.6})$$

that is, C and C_1 are harmonic related with respect to $|Z_1|$ [and so are B and B_1 , as in (A.4)]. Again, this is valid only for the rectangular coordinate system.

Case IV: Two Arbitrary Points B_2 and B_3 on the Locus Circle for Z_{in}

For the fourth case, given two arbitrary points B_2 and B_3 on the locus circle for Z_{in} , find the characteristic impedance Z_1 (real).

Solution: Draw a circle through B_2 and B_3 with its center on the real axis (we assume Z_1 is real). As shown in Fig. A.4, this circle intersects the real axis at the extreme impedances Z_l and Z_h . Next, draw a tangent to the above circle from the origin O . The “touching point” is denoted “T.” Finally, draw a circle with center at O and radius OT . It intersects the real axis at point T_1 . OT_1 is the desired characteristic impedance Z_1 .

Proof: Again, from the “Power of the Circle” [123], $Z_l \cdot Z_h = (OT)^2 = Z_1^2$ (Z_1 is real), which proves our claim. This result is valid only in the rectangular

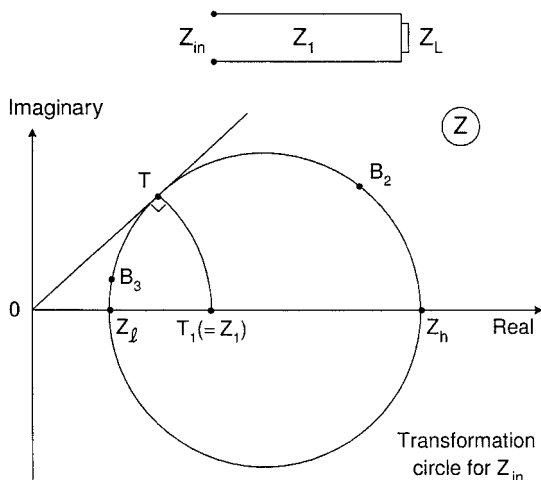


Fig. A.4 Case IV: Given two arbitrary points B_2 and B_3 , the transformation circle for Z_{in} can be drawn immediately with center on the real axis assuming that the characteristic impedance Z_1 is real. Next draw a tangent from the origin O touching the transformation circle at point T . The length of $OT = |Z_1|$. This approach is only valid in the rectangular coordinate system unless actual numbers are read in the Smith chart.

coordinate system unless we read the actual numbers in the Smith chart. See Problem B.2.

A.3 WHERE IS Z_{in} LOCATED ON THE TRANSFORMATION CIRCLE? DETERMINATION OF THE POSITION CIRCLES

So far we have concentrated merely on determining the transformation circle for Z_{in} . In this section we discuss how to determine where on that circle Z_{in} is located. Since this subject is thoroughly described in reference 123, only a brief overview of the approach is given.

First, draw a circle through Z_L and Z_1 with its center on the imaginary axis as showed in Fig. A.5. Draw the tangent to this circle at point Z_1 . Next draw another line through Z_1 intersecting the first one at the angle $2\beta_1 d_1$ (clockwise direction) where β_1 is the propagation constant of the line and d_1 is the length. Finally, draw the position circle through Z_1 having the second line as a tangent and its center located on the imaginary axis. The intersection between this position circle and the transformation circle determines Z_{in} as indicated in Fig. A.5.

PROBLEMS

A.1 Given a load impedance $Z_L = 100 + j50$ ohms, find the characteristic impedance Z_1 and the shortest cable length d_1 that produces an input impedance Z_{in} equal to

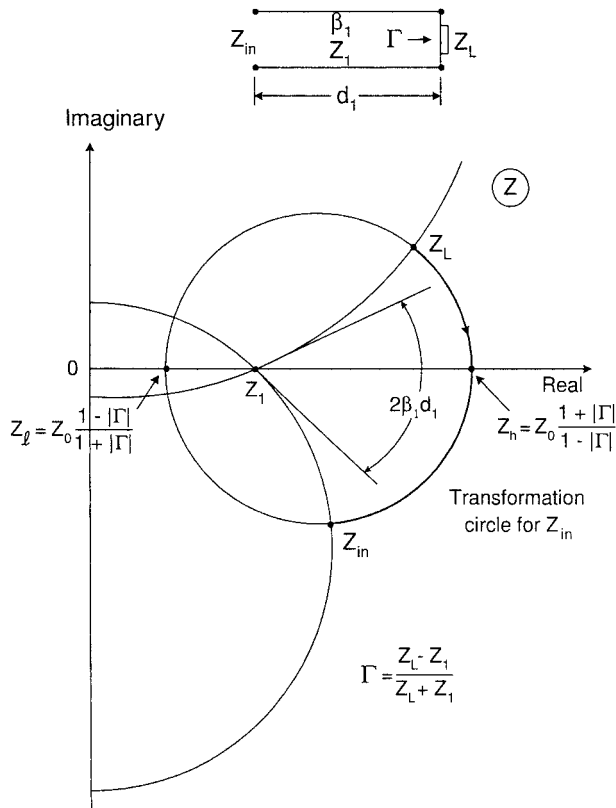


Fig. A.5 Determination of Z_{in} on the transformation circle. Draw a circle with center on the imaginary axis and going through Z_L and Z_1 as shown. Draw the tangent to this circle at point Z_1 . Draw a line, making an angle $2\beta_1 d_1$ with the above tangent (clockwise direction). Draw the position circle with its center on the imaginary position axis and going through Z_1 and with the second line as a tangent. The intersection between this position circle and the transformation circle determines Z_{in} as shown.

- (a) 50 ohms
- (b) 100 ohms
- (c) 150 ohms

You are of course welcome to use a graphical approach. The brightest student usually knows what goes on in 10 minutes or less. If you use a computational approach, be prepared to spend a “bit” more time. And don’t get too frustrated about case (b) or a few other values in that neighborhood.

Computer power is wonderful. However, it is a supplement, not a substitute, for brain power.